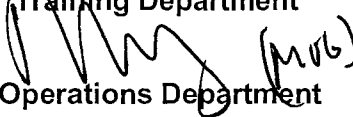


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	EOP Admin. (CVCS)		
TASK:	Determine the CSD boron concentration and calculate the requisite boron addition for a natural circulation cooldown.		
TASK NUMBER:	1150050501		
JPM NUMBER:	14-01 NRC RO Admin A1-2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.43
		IMPORTANCE FACTOR:	4.1
APPLICABILITY:		RO	SRO
EO	☐	RO	☒
STA	☐	SRO	☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	^{21/15} 2 ^{23/15} 1-EOP-TRIP-4, Natural Circulation Cooldown, Rev. 22 All rev checked 7/16/8 S2.RE-RA.ZZ-0016, Unit 2 Curve Book, Rev. 7 S2.RE-RA.ZZ-0012, Figures, Rev. 207		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	15 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	9-8-15
Validated By:	J Page / M Spencer SME or Instructor	Date:	9-25-15 / 9-28-15
Approved By:	 Training Department	Date:	10-7-15
Approved By:	 Operations Department	Date:	10-9-15
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: EOP Admin. (CVCS)

TASK: Determine the CSD boron concentration and calculate the requisite boron addition for a natural circulation cooldown.

TASK NUMBER: 1150050501

INITIAL CONDITIONS:

1. Unit 2 was tripped from full power when off-site power was lost. All vital busses are powered from their respective EDG. The CRS has transitioned to 2-EOP-TRIP-4 in **ANTICIPATION** of beginning a natural circulation cooldown.
2. Core Burnup is 5,000 EFPH.
3. Current RCS boron concentration is 1,050 ppm.
4. RCS Tc's are 547°F and stable.

INITIATING CUE:

You are the RO. Perform Step 4 to determine the required CSD Boron Concentration, then determine the boration required to achieve CSD boron concentration.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Provides the CSD boron concentration in ppm (1950-1970 ppm)
2. Provides the boration required in gallons to achieve CSD boron concentration (11,370-11,646 gallons)

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: EOP Admin. (CVCS)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide a copy of 2-EOP-TRIP-4, Natural Circulation Cooldown, Sheet 1. Provide clean copies of: S2.RE-RA.ZZ-0016, Rev. 7, Curve Book S2.RE-RA.ZZ-0012, Rev. 207, Figures	Reviews procedure.		
		START TIME:			
*	TRIP-4 Step 4	Using C_B from Figure 20A of "Reactor Engineering Manual, determine required boron concentration for Cold Shutdown (All Rods In, $K_{eff}=0.95$) as follows: $\frac{\quad}{C_B} \text{ PPM} + 50 \text{ ppm} = \frac{\quad}{\text{Req } C_B} \text{ ppm}$	Locates Figure 20A in Unit 2 Curve Book Note: Table A on page 111 are the data points for Figure 20A. The exact number for 5,000 EFPH is <u>1914 ppm</u> . Allowable graph interpolation is 1900-1920 ppm . Adds 50 ppm. The exact number is <u>1964 ppm</u> . Allowable range is 1950-1970 ppm NOTE: Using the $k_{eff}=0.99$ line instead of the correct $k_{eff}=0.95$ line will result in an incorrect boron concentration of $1452+50=1502 \text{ ppm}$		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: EOP Admin. (CVCS)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*		Determine the boration required to achieve CSD boron concentration.	Using Figure 103 of S2.RE-RA.ZZ-0012, Tables, and the formula on that page, determines 11,563 gallons of boron must be added. Allowable range of answer is 11,370-11,646 gallons of boron required to achieve CSD concentration. May either pick number off Figure 103, Boron Addition Graph, or use calculation provided on same figure. NOTE: Since the boration occurs BEFORE any cooldown, the correction factor for temperature on Figure 105 is not required, since it is only for temperatures below 547.		
			IF the extra 50 ppm is not added, the incorrect result will be 10,876 gallons.		
			Provides two numbers: 1. CSD boron concentration in ppm. 2. Boration required in gallons to achieve CSD boron concentration. When these two numbers have been given to the evaluator, state JPM is complete.		
		STOP TIME:			

Terminating Cue: None

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- Q P 1. Task description and number, JPM description and number are identified.
- Q P 2. Knowledge and Abilities (K/A) references are included.
- Q P 3. Performance location specified. (in-plant, control room, or simulator)
- Q P 4. Initial setup conditions are identified.
- Q P 5. Initiating and terminating Cues are properly identified.
- Q P 6. Task standards identified and verified by SME review.
- Q P 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- Q P 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 22 Date 9-25-15
- Q P 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- N/A 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- N/A 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] Date: 9-25-15

SME/Instructor: Michael Spencer Date: 9-28-15

SME/Instructor: _____ Date: _____

INITIAL CONDITIONS:

1. Unit 2 was tripped from full power when off-site power was lost. All vital busses are powered from their respective EDG. The CRS has transitioned to 2-EOP-TRIP-4 in **ANTICIPATION** of beginning a natural circulation cooldown.
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INITIATING CUE:

You are the RO. Perform Step 4 to determine the required CSD Boron Concentration, then determine the boration required to achieve CSD boron concentration.